

Reminders

Upcoming due dates

Wed Feb 18 Data checkpoint

Fri Feb 20 D6

Machine Learning

Data Science in Practice

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Hello! My COGS 108 group is doing research on clothing consumption and disposal behaviors in college students. We would appreciate it if you took a few minutes to fill out our survey. It will be anonymous, and it is only 6 multiple choice questions!

<https://forms.gle/WHr9RQT6SwZ8GxXXA>



SYN 100 - Campus & Climate Change

Interested in the Climate Impact of Using AI?

Help us quantify the impact by filling out the survey below!





model assessment

Squared Error

Computationally simpler

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n \left(Y_i - \hat{Y}_i \right)^2$$

Same units as y

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

A few outliers can lead to a big increase in squared errors... even if all the other predictions are pretty good

Alternatively

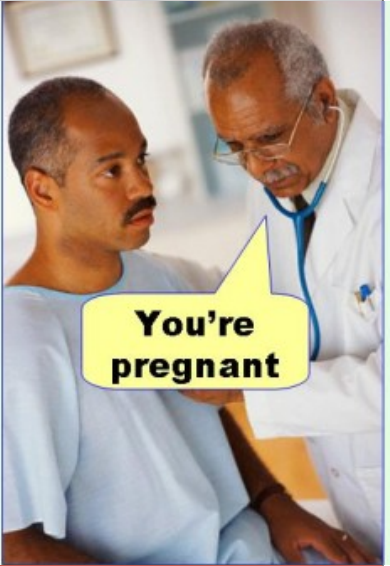
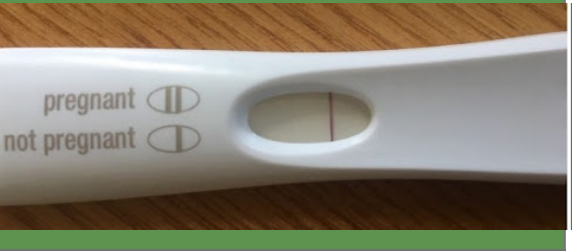
$$\text{MAE} = \frac{\sum_{i=1}^n |y_i - x_i|}{n} = \frac{\sum_{i=1}^n |e_i|}{n}.$$

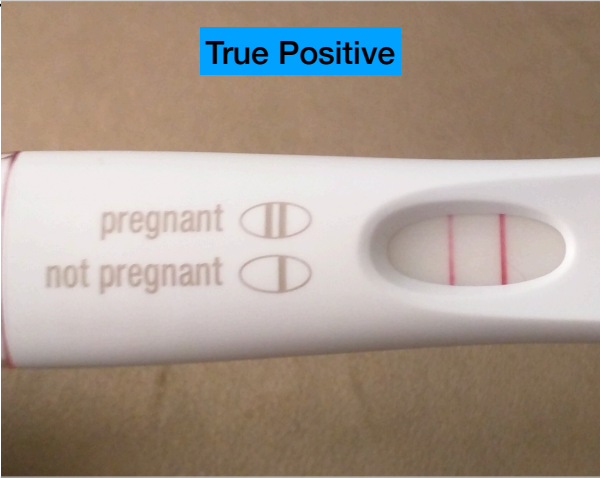
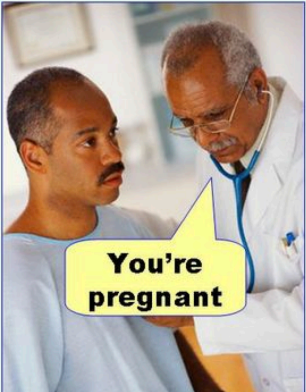
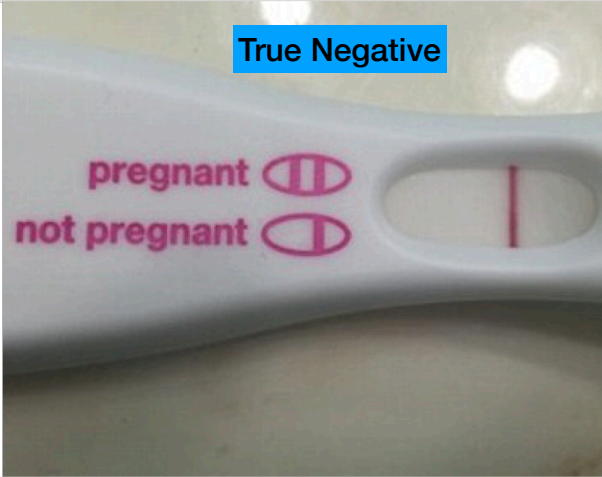
If you care more about most errors, and less about the outliers

$$\text{Accuracy} = \frac{\# \text{ of samples predicted correctly}}{\# \text{ of samples predicted}} * 100$$

Accuracy can mislead

- If classes are imbalanced, what would “chance performance” be?
- The classic example: detect cancer
 - 1/1000 actually have cancer
 - your prediction algorithm misdiagnoses 1% of healthy people
 - Do the math... your algorithm tells 10 people who are healthy they are sick for every 1 person who is actually sick

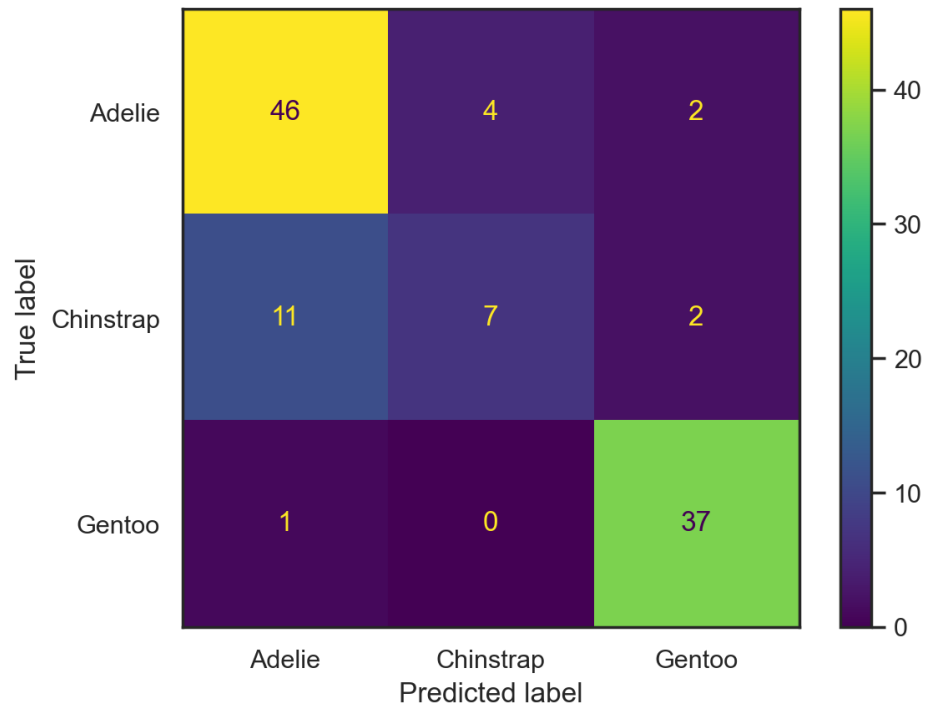
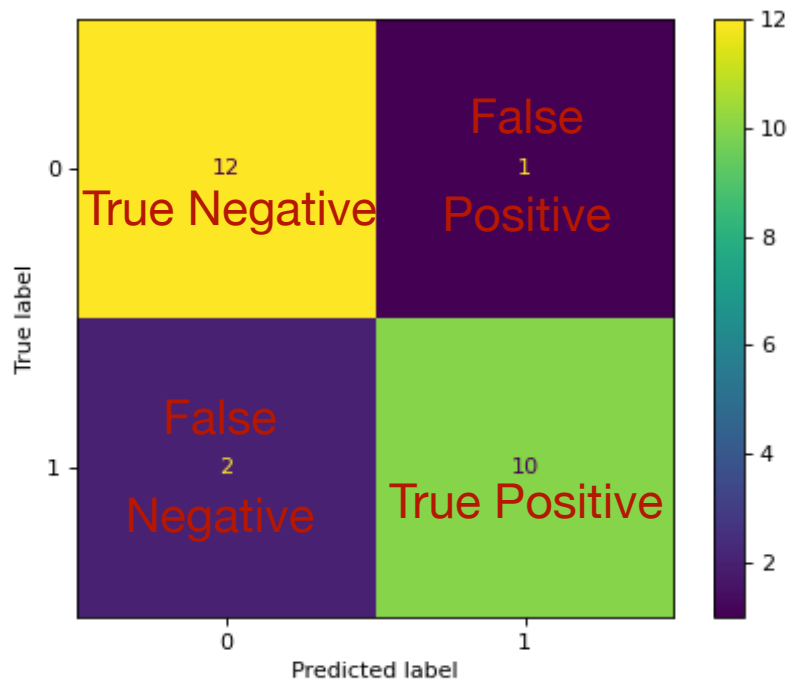
		Actual	
		Positive	Negative
Predicted	Positive	True Positive (TP) 	False Positive (FP) 
	Negative	False Negative (FN) 	True Negative (TN) 

	Predicted +	Predicted -
Truly +	True Positive 	Type II error (false negative) 
Truly -	Type I error (false positive) 	True Negative 

confusion matrix

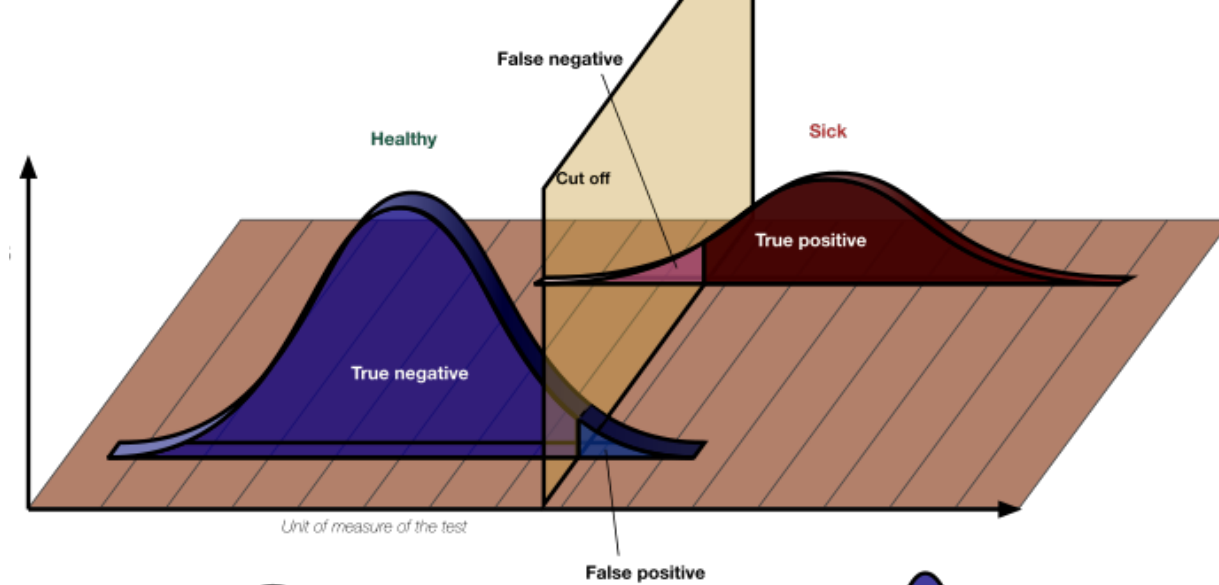
		Actual	
		Positive	Negative
Predicted	Positive	True Positive (TP)	False Positive (FP)
	Negative	False Negative (FN)	True Negative (TN)

Confusion matrix



Accuracy	What % were predicted correctly?
Recall a.k.a. Sensitivity	Of those that <i>were</i> positives , what % were predicted to be positive?
Specificity	Of those that were negatives , what % were predicted to be negative?
Precision a.k.a. PPV	Of those that we predicted positive, what % were positives ?

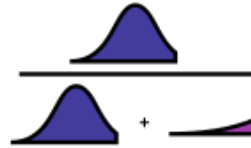
categorical variable prediction



PPV =
aka recall



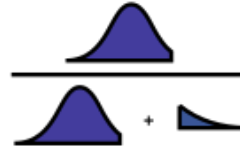
NPV =



Sensitivity =
aka recall



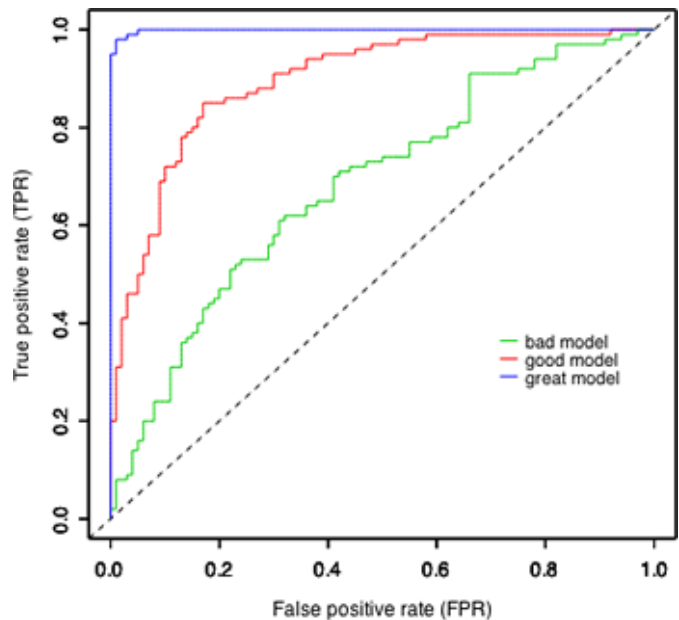
Specificity =



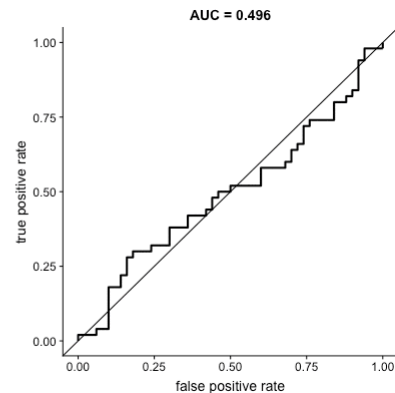
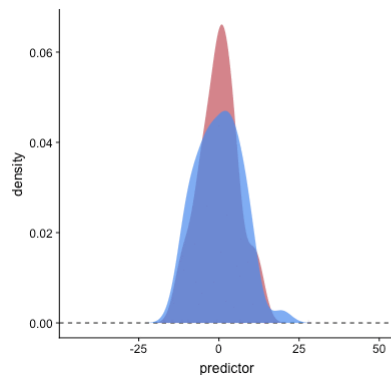
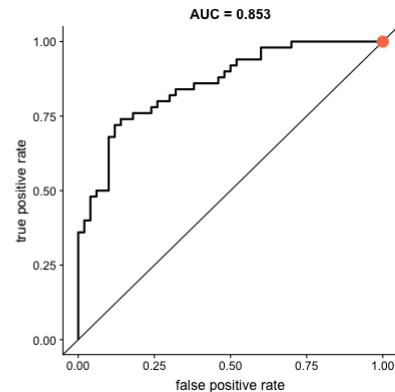
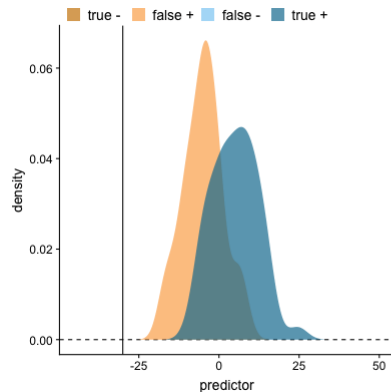
		Actual	
		Positive	Negative
Predicted	Positive	True Positive (TP)	False Positive (FP)
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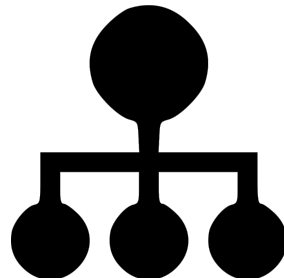
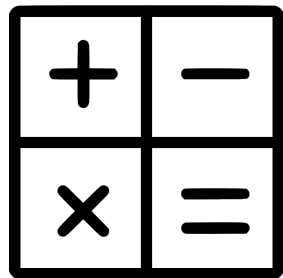
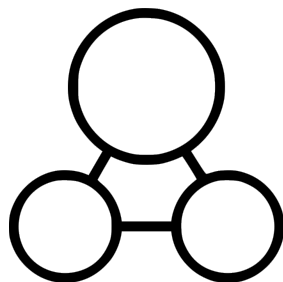
Receiver Operating Characteristic

ROC-AUC



The value can range from 0 to 1. However AUC score of a random classifier for balanced data is 0.5





model selection

OUR GOAL IS TO PREVENT OVERFITTING!!!!

Selecting the right model form

Things we can select

- Try different algorithms: k-NN, SVM-linear, Neural network, ...
- For each algorithm try a variety of hyperparameters: $k=1,3,5,7,9\dots$
- For each algorithm try different features: all possible subsets of n variables
- Or any subset of the above!

Selecting the right model form

The most common thing is selecting hyperparameters

- Hyperparameter selection
 - k-NN: $k=1,3,5,7,9,11,\dots$
 - You pick k based in part on the dataset. 300 samples and 10 categories? Don't go very high. 300k samples and 2 categories? You can go very big k !
 - SVM: $C=0.0001, 0.001, 0.01, 0.1, 1, 10, 100, 1000, 10000$
 - Your dataset may inform here too... noisy problem, you need soft C values (<1)
 - Decision tree: $\text{max_depth} = 2, 3, 4, 5, 6, 7\dots$
 - Note that number of tree nodes are $2^{\text{max_depth}}$ so this is an exponential progression in complexity just like in SVM above

<https://arstechnica.com/health/2023/11/ai-with-90-error-rate-forces-elderly-out-of-rehab-nursing-homes-suit-claims/>

Preventing overfitting is important!

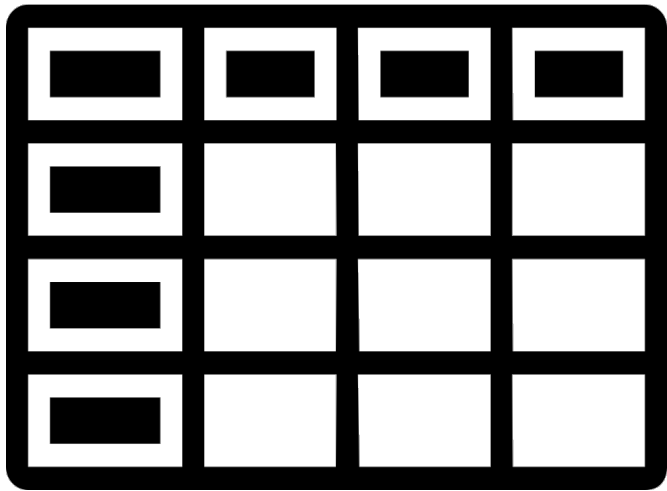
When AI models make errors it is typically because the dataset is a bad sample, leading to increased likelihood of overfitting patterns that are particular to this bad sample.

Generically there are two issues at play here.

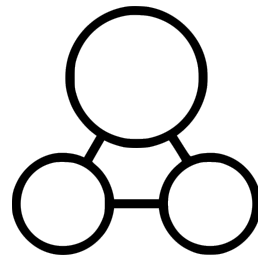
Bad procedures (or carelessness) led to a bad model

Then ethical failings led to this bad model being put into service and kept in service





BIGGER
datasets



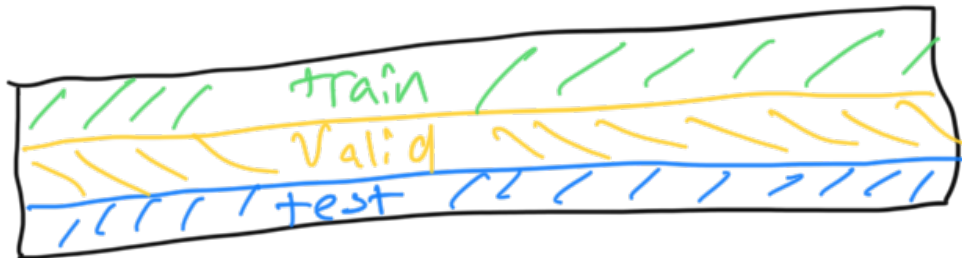
SIMPLER
models

①

All the data

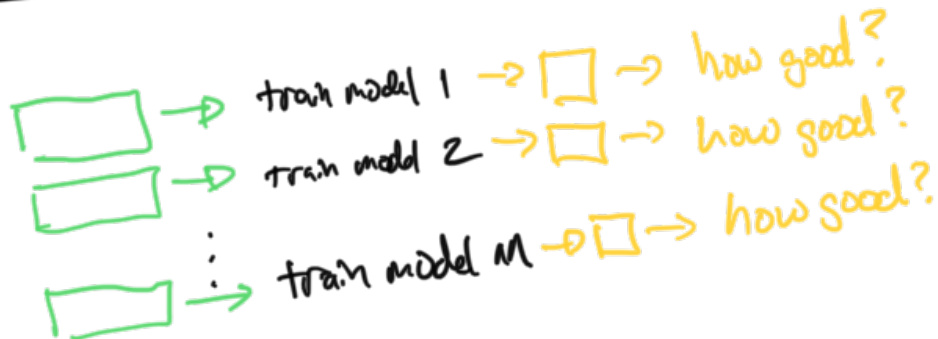
randomly shuffle
rows of dataset

②



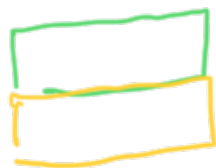
Split into
training, validation
and test sets

③



if you have M
models to try do this
once for each

④



train
best
model
above

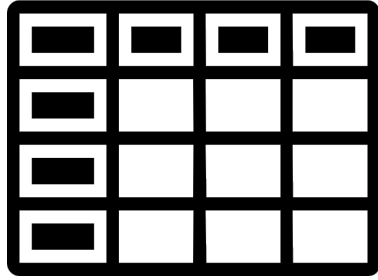


Estimate
of how
good it will
be in future

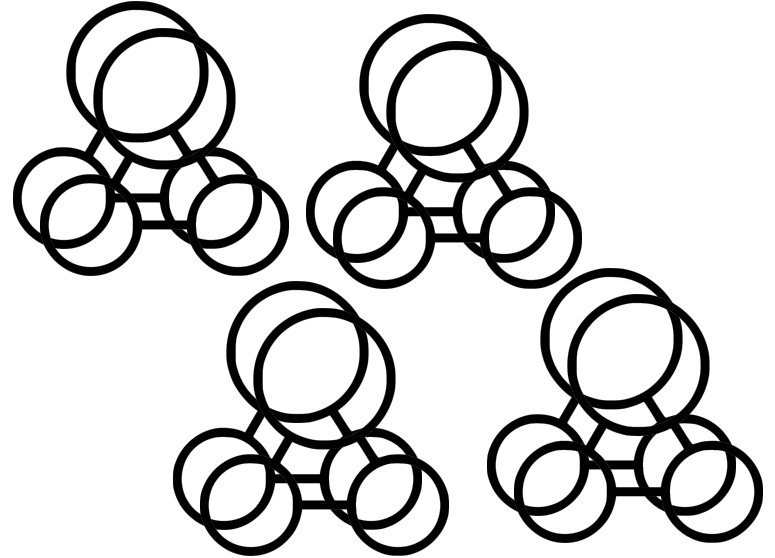
Pick best model from
validation set performance
train it on $T+V$ then test
it on $Test\ set$

Model, feature, and hyperparameter selection

- Needs: Don't leak information from training/selection process into the test set!
- Trade-offs: Usually not enough data to have completely separate train, validation, test sets. Which one do we prioritize?
 - Low training data -> bad fit
 - Low validation data -> bad selection of model/feature/hparam
 - Low test data -> poor estimate of generalization performance

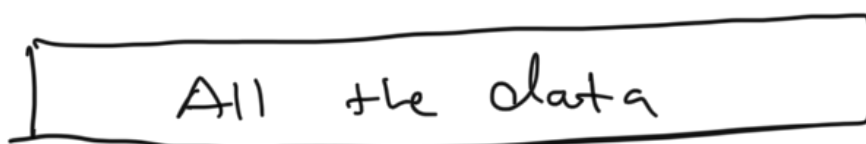


SMALLER
datasets



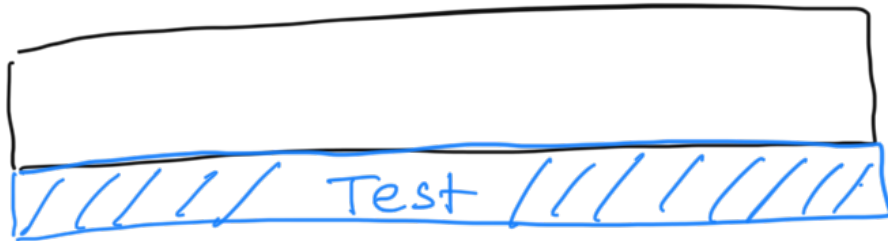
COMPLEX
models

①

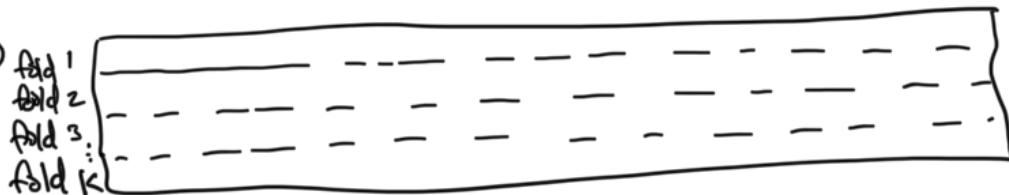


randomly shuffle
rows of dataset
split off a
test set

②

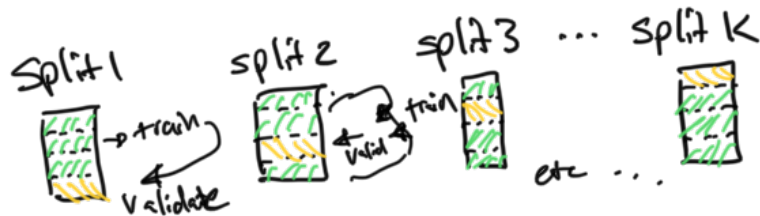


③



split remaining data
into K folds with
even(ish) amounts of data
w/ some(ish) distributions

④



Every split gets its turn as the validation
set, use all other splits as training.
Validation performance is the mean
across splits

⑤

Do steps ③ + ④ M times to test M models,
pick best validation performance model



trained
best
model



Estimated
performance
in future



Prediction Approach



[https://forms.gle/
ubN7LFy3FdbSmcEi6](https://forms.gle/ubN7LFy3FdbSmcEi6)